

Chapter 8 – Package Management with APT

Objective

Ubuntu uses the **Advanced Package Tool (APT)** to install, update, upgrade, and remove software. Instead of downloading applications manually, APT retrieves packages from trusted software repositories.

In this chapter, you will learn how to:

- Understand APT.
 - Update package information.
 - Upgrade installed software.
 - Install new packages.
 - Remove packages.
 - Search for software.
 - Clean package caches.
 - View installed packages.
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8.1 What is APT?

APT (Advanced Package Tool) is Ubuntu's default package management system.

APT simplifies software management by automatically:

- Downloading packages
- Resolving dependencies
- Installing software
- Updating packages
- Removing applications
- Managing software repositories

Instead of manually downloading software from websites, administrators can install applications using a single command.

8.2 Understanding Ubuntu Repositories

Ubuntu stores software in **repositories**, which are trusted servers maintained by Canonical and the Ubuntu community.

When APT installs software, it downloads packages directly from these repositories.

Typical repository categories include:

- Main
- Universe
- Restricted
- Multiverse

These repositories contain thousands of software packages.

8.3 Update Package Information

Refresh the package index before installing software.

```
sudo apt update
```

Purpose

- Downloads the latest package information.
 - Checks for newer package versions.
 - Does **not** install updates.
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8.4 Upgrade Installed Packages

Install available software updates.

```
sudo apt upgrade -y
```

Purpose

- Upgrades installed packages.
 - Installs security patches.
 - Keeps the operating system up to date.
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8.5 Install Software

Install a package using:

```
sudo apt install <package-name>
```

Example:

```
sudo apt install git
```

Install multiple packages:

```
sudo apt install git curl wget
```

8.6 Search for Packages

Search available packages.

```
apt search nginx
```

Example output:

nginx

nginx-common

nginx-core

This helps locate software before installation.

8.7 View Installed Packages

Display installed packages.

```
apt list --installed
```

Or search for a specific package.

```
dpkg -l | grep git
```

Example:

```
ii git
```

8.8 Remove Packages

Remove software while keeping configuration files.

```
sudo apt remove nginx
```

To completely remove software and its configuration files:

```
sudo apt purge nginx
```

8.9 Remove Unused Packages

Over time, Ubuntu accumulates unnecessary dependencies.

Clean them using:

```
sudo apt autoremove
```

This removes packages that are no longer required.

8.10 Clean Package Cache

Downloaded package files remain stored on disk.

Remove them with:

`sudo apt clean`

To remove only obsolete packages:

`sudo apt autoclean`

Cleaning the cache helps free disk space.

8.11 Useful Package Management Commands

Command	Purpose
<code>sudo apt update</code>	Refresh package list
<code>sudo apt upgrade</code>	Install available updates
<code>sudo apt install <package></code>	Install software
<code>sudo apt remove <package></code>	Remove software
<code>sudo apt purge <package></code>	Remove software and configuration
<code>sudo apt autoremove</code>	Remove unused dependencies
<code>sudo apt clean</code>	Clear package cache
<code>apt search <package></code>	Search for packages
<code>apt list --installed</code>	List installed packages

8.12 Best Practices

- Run `sudo apt update` before installing software.
 - Regularly apply security updates.
 - Remove unused packages.
 - Use official Ubuntu repositories whenever possible.
 - Avoid downloading software from untrusted sources.
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Chapter Summary

In this chapter, you learned how to:

- Understand APT.
- Install software.
- Update Ubuntu.
- Upgrade installed packages.
- Remove applications.
- Search for packages.
- Maintain a clean package cache.

APT is one of the most important tools for managing an Ubuntu Server.

One-Line Summary

APT is Ubuntu's package management system, providing a secure and efficient way to install, update, remove, and maintain software from trusted repositories.